SLE-4: Final Viva & Architecture Decision Record (ADR)

02AML204 – Introduction to Artificial Intelligence

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PART A: ARCHITECTURE DECISION RECORD (ADR)

Title: Use of A* Search with Manhattan Distance Heuristic

Status: Proposed (confirm against the implementation)

1. Context

The project involves finding a path between a starting position and a goal in a grid-based search environment. The search method must explore possible states while avoiding unnecessary work where possible. Different search algorithms make different trade-offs between completeness, path quality, and the amount of exploration required. A suitable heuristic can help guide the search toward the goal. The decision therefore concerns choosing a search strategy and heuristic that fit this type of problem.

2. Decision

Use the A* search algorithm with Manhattan distance as the heuristic for a grid in which movement is restricted to horizontal and vertical steps. A* ranks candidate states using the path cost so far and an estimate of the remaining distance to the goal.

3. Alternatives Considered

- **Breadth-First Search (BFS):** useful for finding a shortest path when all moves have equal cost, but it does not use goal-distance guidance.
- **Depth-First Search (DFS):** can use less memory in some cases, but it does not guarantee a shortest path.
- **Greedy Best-First Search:** uses a heuristic to move toward the goal, but can ignore the cost already paid and may return a non-optimal path.

4. Rationale / Justification

A* combines the cost already incurred, g(n), with an estimate of the remaining cost, h(n), using f(n) = g(n) + h(n). Manhattan distance is appropriate for a four-directional grid because it estimates the number of horizontal and vertical steps to the goal. This gives the search useful direction while still accounting for the path cost so far. Compared with uninformed search, this approach can avoid exploring some irrelevant areas, depending on the grid and obstacle layout. Compared with Greedy Best-First Search, A* considers both progress and estimated remaining distance. The design can keep state tracking, search logic, and result presentation in separate responsibilities if that matches the project's C4 architecture. The SLE-2 profiling report should be used to verify the actual runtime, states explored, and memory behaviour; no measured values are stated here because the profiling results were not provided.

5. Consequences

Positive:

- Provides a systematic search strategy that uses both path cost and goal-distance estimate.
- With suitable conditions and an admissible heuristic, A* can find an optimal path.
- The heuristic is simple to calculate for four-directional grid movement.

Negative / Trade-offs:

- A* may consume substantial memory because it stores discovered states and candidate states.
- Performance depends on the map, obstacles, and heuristic quality.
- Manhattan distance is not suitable without adjustment when movement rules allow diagonal steps with different costs.

6. References

- SLE-2 profiling and comparison report (project-specific results to be added).
- SLE-3 C4 architecture diagrams (project-specific components to be confirmed).

PART B: SHORT SUMMARY OF MY SLE JOURNEY

In SLE-1, the work focused on implementing the AI agent/search solution and documenting the implementation. In SLE-2, the solution was profiled and compared using performance measurements. In SLE-3, the system structure was

represented through C4 architecture diagrams at different levels. This ADR connects the implementation, performance evaluation, and architecture by recording the reasoning behind the search algorithm and heuristic choice. The final justification should be checked against the actual code, profiling data, and diagrams.

PART C: AI CONTRIBUTION NOTE

AI tools were used to help organise and draft this ADR and to prepare possible viva questions. I am responsible for checking the report against my own implementation, confirming the alternatives, adding actual SLE-2 profiling results, and ensuring that the SLE-3 architecture description matches my diagrams. Any AI-generated content will be reviewed and edited to reflect my actual work.

PART D: VIVA PREPARATION NOTES

- A* evaluates a state using f(n) = g(n) + h(n).
- Manhattan distance sums horizontal and vertical distance to the goal.
- BFS does not use a heuristic; DFS does not guarantee a shortest path.
- A* can require significant memory on large search spaces.
- Be ready to explain the actual SLE-2 measurements and the SLE-3 Container diagram.

Note: This report uses the A* with Manhattan-distance example from the supplied guideline because the user's actual project details and profiling results have not yet been shared. Confirm that this matches your project before submitting; replace or revise any details that do not.